INCH-POUND

MIL-B-70731A(AR)
02 November 1993
SUPERSEDING
MIL-B-70731 (AR)
10 July 1988

MILITARY SPECIFICATION

BATTERY, PRIMARY RESERVE, AMMONIA 5 MILLIAMPERE HOUR CAPACITY, FOR MINE, ANTI-PERSONNEL: M86

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, manufacture and assembly of the battery used in the molded assembly M86, Mine, Anti-personel. (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (see 6.2)

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1345

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STANDARDS

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in effect in the solicitation.

DRAWINGS (See 6.7)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12929222 - Battery, Assembly

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts and assemblies. The parts and assemblies shall comply with all requirements specified on dwg. 12929222 and associated drawings and with all requirements specified in applicable specifications.

3.3 Ampule

3.3.1 Seal The ampule shall not show evidence of leakage when tested in accordance with 4.5.1.1.

3.3.2 Environmental. The ampule shall be capable of withstanding exposure to the environments listed herein.

3.3.2.1 Temperature cycling. The ampule shall be capable of withstanding temperature cycling from -65°F + 5°F to + 170°F + 5°F - 0°F when tested as specified in 4.5.1.2.3

3.3.2.2 Time and temperature to fragmentation. The ampule, after an initial stabilization at 160°F for at least one hour, shall be exposed to an increasing temperature of 5°F per 15 minute period. The ampule shall not be fragmentized at any temperature of 170°F or below."

3.3.3 Ammonia composition.

a. The ammonia used to fill the ampule shall be anhydrous and of 99.98% minimum purity.

b. The filled and sealed ampule shall not contain more than .10% water.

3.3.4 Ammonia weight. The weight of ammonia in a sealed ampule shall be 85 + 10 milligrams.

3.4 Cathode.

3.4.1 Moisture content. The cathode material shall not average more than 1.0% of water. No sample shall have a moisture content greater than 1.5%.

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3.4.2 Weight. The cathode weight shall be 78 ± 6.0 milligrams.

3.5 Battery primary reserve.

3.5.1 Case seal and pre-initiation voltage. The battery shall be sealed to prevent leakage and not allow an output voltage in excess of 0.01 of a volt.

3.5.2 Activation. The battery shall be initiated and activated in accordance with 4.5.9.2.

3.5.3 Rise time. The battery output voltage shall rise to 3.3 + 1.9 volts within ten (10) seconds when tested in accordance with 4.5.9.3 for a test temperature of +120°F + 5°F to +32°F + 5°F. For temperature ranging from +32°F + 5°F to -20°F -5°F, the voltage shall rise to 3.3 + 1.9 volts within ten (10) minutes. During the rise time, the output voltage shall not regress below 2.2 volts after attaining 2.7 volts in the first 14% of timing but reaches and maintains at least 2.2 volts between 14% and 100% shall be classed defective.

3.5.4 Time capacity. The battery shall be activated to the requirements of 3.5.2, meet the rise time requirements of 3.5.3 and have an output voltage of no less than 2.7 volts at +120° +5°F, no less than 3.5 volts at -20° -5°F, or no less than 2.8 volts at Ambient for four (4) hours of continuous operation when tested as specified in 4.5.9.4.

3.5.5 Test temperature. The battery shall be capable of meeting all specified requirements at any temperature from -20°F -5°F to +120°F +5°F.

3.6 Environmental and mechanical test.

3.6.1 Thermal shock. The battery shall be capable of withstanding changes in temperature when tested in accordance with 4.5.10.1.

3.6.2 Temperature cycling. The battery shall be capable of withstanding changes in temperature when tested in accordance with 4.5.10.2.

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3.6.3 Transportation vibration. The battery shall be capable of withstanding mechanical shocks when tested in accordance with 4.5.10.3.

3.6.4 Drop test. The battery shall be capable of withstanding shock when tested as specified in 4.5.10.4.

3.7 Repair. Any module failing to comply with all the test requirements and undergoes repair shall be tested specified in 4.5.11.

3.8 Test temperature. Unless specified otherwise, all testing shall be performed at a temperature of 25°C ± 5°C.

3.9 Data requirements. Data shall be recorded and retained for all tests required by this specification and as specified in 4.5.12.

3.10 Process controls. All process controls essential for battery manufacturer as deemed necessary by the Government to assure a reliable end item shall be submitted and maintained in accordance with 4.5.13.

3.11 Qualification. The battery shall be requalified if:

a. The contractor's manufacturing techniques or process controls which affect the form, fit, function or internal geometry of the battery are revised by either the contractor or one of his sub-contractors supplying parts or assemblies.

b. The contractor relocates his assembly facilities.

c. The contractor uses a new supplier for major sub-assemblies.

d. A change in material used for battery production or assembly.

Qualification requirements shall be determined by the Government for each change as specified in 4.4.3.14.

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3.12 Critical safety item process control. A process control document shall be prepared to fully describe the entire process and controls for all characteristics containing the CSI notation. This document shall be submitted to the Government for approval.

3.13 Workmanship. All mechanical assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agents. All markings and stampings shall be free of harmful cleaning and soldering agents that may be injurious to the assembly. All markings and stampings, either component or assembly shall be displayed and visibly defined. The battery shall be manufactured and processed in such a manner as to be uniform in quality and free from defects that will affect life, configuration and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provision. Unless otherwise specified herein, the provision of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define qualtiy assurance terms used herein.

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4.2 Classification of inspections. The inspection requirements specfied herein are classified as follows:

a. First article inspection.

b. Quality conformance inspections.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated:

PART DESCRIPTION	DRAWING NO.	QUANTITY
Ampule and Case Assembly	12929221	25
Bulkhead	9298627	25
Ampule, Glass	12992220	75
Case	9298629	25
Separator	9298631-1	25
Separator	9298631-2	25
Cover and Collar Assembly	12929218	25
Anode No. 1	9298633-1	25
Anode No. 2	9298633-2	25
Collar	12929215	25
Cover Assembly	9298636	25
Cathode	9298640	75
Cover and Cell Assembly	12929219	25
Cell Cup	12929216	25
Battery, Primary Reserve, Assembly	1292922	336
Support, Ampule	9298650	25

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4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein and the battery assembly test. Battery assemblies listed in Table I shall be tested as described below (see Figure 1):

a. 36 batteries shall be subjected to the four (4) environmental tests in the sequence shown. After the environmental testing, the batteries shall be subjected to the electrical test which includes activation (4.5.9.2), rise time (4.5.9.3) and time capacity (4.5.9.4). However, the batteries shall not require rotation (spin) during activation. 12 each batteries shall be tested at the temperature listed in Table II of 4.4.3.7.3. If the test sample contains one (1) reject for activation, rise time or time discharge requirements, a second sample of 36 batteries shall be tested. The second sample shall have passed the above criteria providing there are zero (0) rejects in the sample. The test shall be rejected if the first sample or the combined sample contain two (2) or more rejects for activation, rise time or discharge requirements of 4.4.3.7.

b. 225 shall be subjected to activation, rise time and time capacity. 75 batteries shall be tested at the three (3) temperatures listed in Table II of 4.4.3.7.3. Three (3) rejects shall be allowed for activation and rise time requirements. Three (3) rejects shall also be allowed for time capacity requirements per 4.5.9.4.

4.3.3 Rejection. See MIL-A-48078. The First Article sample shall be rejected if any part, sub-assembly or assembly fails to comply with any of the examinations or test specified in this specification or any of the criteria specified in 4.3.2. above.

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Table I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE Battery, Primary Reserve Ammonia 5 Milliampere Hour Capacity	SHEET 1 OF 4	DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Ampule and Case Assembly (Dwg. No. 12929221)			
	Presence of ammonia in ampule	25	3.2	4.5.8
	Bulkhead to case force test	25	3.2	4.5.3
	Bulkhead (Dwg. No. 9298627)			
	Examination for defects	25	3.2	4.4.2.2
	Ampule Assy, Glass (Dwg. 12929220)			
	Examination for defects	25	3.2	4.4.2.3
	Seal	25	3.3.1	4.5.1.1
	Ammonia composition	25	3.2	4.5.1.3
	Weight	25	3.2	4.5.1.4
	Temperature Cycling (Short Term)	25	3.3.2.1	4.5.1.2.3

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used

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Table I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE Battery, Primary Reserve Ammonia 5 Milliampere Hour Capacity	SHEET 2 OF 4	DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Case (Dwg. 9298629) Examination for defects	25	3.2	4.4.2.4
	Separator (Dwg. 9298631-1) Salt content	25	3.2	4.5.7
	Separator (Dwg. 9298631-2) Salt content	25	3.2	4.5.7
	Cover and collar assembly (Dwg. 12929218) Weld strength Vacuum pull test	25 25	3.2 3.2	4.5.6.1 4.5.6.2
	Anode No. 1 (Dwg. 9298633-1 Examination for defects	25	3.2	4.4.2.6
	Anode No. 2 (Dwg. 9298633-2) Examination for defects	25	3.2	4.4.2.7

NOTES:

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Table I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE Battery, Primary Reserve Ammonia 5 Milliampere Hour Capacity	SHEET 3 OF 4	DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY	
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Collar (Dwg. 12929215) Examination for defects	25	3.2	4.4.2.8
	Cover assembly (Dwg. 9298636) Examination for defects	25	3.2	4.4.2.9
	Cathode (Dwg. 9298640) Moisture Content Weight Composition	25 25 25	3.4.1 3.4.2 3.2	4.5.2.1 4.5.2.2 4.5.2.3
	Cover and Cell Assembly (Dwg. 12929219) Weld Test Vacuum Full Test	25 25	3.2 3.2	4.5.5.1 4.5.5.2
	Cell Cup (Dwg. 12929216) Examination for Defects	25	3.2	4.4.2.11

NOTES:

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Table I. First article inspection

CLASSIFICATION OF CHARACTERISTICS MIL-B-70731A(AR)

PARAGRAPH	TITLE Battery, Primary Reserve Ammonia 5 Milliampere Hour Capacity	SHEET 4 OF 4	DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA / REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Support Ampule (Dwg. 9298650)		
	Examination for Defects	25 / 3.2	4.4.2.13
	Hardness test	Note a 336	
	Battery Assembly (Dwg. 9298625)	Note b	
	Thermal Shock	36 / 3.6.1	4.5.10.1/4.3.2.a
	Temperature Cycling	36 / 3.6.2	4.5.10.2/4.3.2.a
	Transportation Vibration	36 / 3.6.3	4.5.10.3/4.3.2.a
	Drop Test	36 / 3.6.4	4.5.10.4/4.3.2.a
	Activation	225 / 3.5.2	4.5.9.2/4.3.2.b
	Rise Time	225 / 3.5.3	4.5.9.3/4.3.2.b
	Time Capacity	225 / 3.5.4	4.5.9.4/4.3.2.c
	Examination for defects	25 / 3.2	4.4.2.12
	Case Seal and Pre-initiation voltage	25 / 3.5.1	4.5.9.1

NOTES:
a. A coupon from the support ampule batch shall be used as the sample.
b. Double sample may apply. See 4.3.2.

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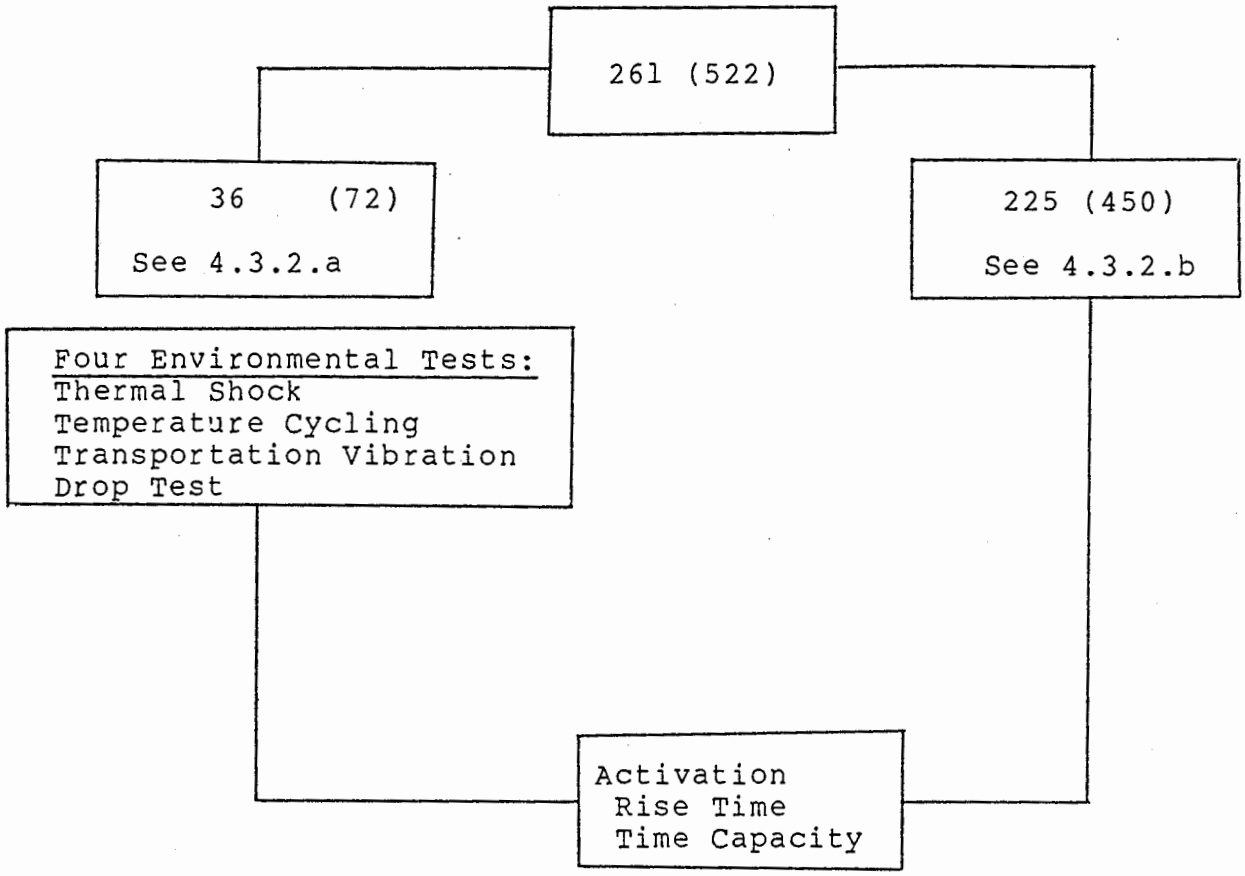


FIGURE 1. First article inspection test sequence for battery assembly.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When specified herein, attributes sampling inspection shall be conducted in accordance with Table I following, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II. Sampling

Lot Size	Inspection Level I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

Special Inspection Levels

Lot Size	S-I	S-II	S-III	S-IV	S-V	S-VI
2 to 15	*	*	*	13	5	3
15 to 50	*	*	32	13	5	3
51 to 150	*	125	32	13	5	3
151 to 500	*	125	32	13	5	3
501 to 3200	1250	125	32	13	13	5
3201 to 35000	1250	125	32	13	13	5
35001 to 500000	1250	125	32	32	20	8
500001 and over	1250	125	32	32	20	8

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.5). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative provides equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Ampule and Case Assembly	1 OF 1	12929221
			NEXT HIGHER ASSEMBLY 12929222

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Presence of ammonia in ampule	Level II	3.2	4.5.8
102	Bulkhead to case force test	Level II	3.2	4.4.3.9/4.5.3
Minor				
201	Poor workmanship	Level III	3.13	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Bulkhead	1 OF 1	9298627
			NEXT HIGHER ASSEMBLY 1292922l

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thickness from outside to bottom of spherical radiusL	Level II	3.2	Gage
Minor				
201	Poor workmanship	Level III	3.13	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Ampule, Glass	1 OF 1	12929220
			NEXT HIGHER ASSEMBLY 12929221

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Seal	Level II	3.3.1	4.5.1.1
102	Temperature cycling	Level II	3.3.2.1	4.5.1.2.3
103	Ammonia composition	Level II	3.3.3	4.5.1.3
104	Depth from base to mid-point at top of shoulder	Level II	3.2	Gage
105	Depth of center point	Level II	3.2	Gage
106	Concentricity of tip	Level II	3.2	Gage
107	Presence of ammonia in ampule	Level II	3.2	Visual
108	Ammonia Weight	4.4.3.8	3.3.4	4.5.1.4
109	Ammonia weight screen	1/	3.3.4	4.5.16
Minor				
201	Poor workmanship	Level III	3.13	Visual

NOTES: 1/ 100% inspection shall be certified by process control.
(a) The lot shall be rejected if one (1) or more samples are classed defective.

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CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Case	1 OF 1	9298629
			NEXT HIGHER ASSEMBLY: 12929221

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside radius at bottom	Level II	3.2	Gage
102	Thickness of bottom	Level II	3.2	Gage
Minor				
201	Flatness of flange	Level III	3.2	Gage
202	Poor workmanship	Level III	3.13	Visual

NOTES:

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MIL-B-70731A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Cover and Collar Assembly	1 OF 1	12929218
			NEXT HIGHER ASSEMBLY: 12929219

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Vacuum pull test	Level II	3.2	4.5.6.2
102	Weld strength	Level II	3.2	4.4.3.4/4.5.6.1
Minor				
201	Poor workmanship	Level III	3.13	Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Anode No. 1	1 OF 1	9298633-1
			NEXT HIGHER ASSEMBLY: 12929219

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Diameter	Level IV	3.2	Gage
202	Poor workmanship	Level IV	3.13	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Anode No. 2	1 OF 1	9298633-2
			NEXT HIGHER ASSEMBLY: 12929219

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Diameter	Level IV	3.2	Gage
202	Poor workmanship	Level IV	3.13	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 — Replaces 1570, 1 Feb 85, which may not be used

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Collar	1 OF 1	12929215
			NEXT HIGHER ASSEMBLY: 12929218

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Web thickness	Level II	3.2	Gage
102	Slot width	1/, Level II	3.2	Gage
Minor				
201	Anode side inside diameter, minimum	Level IV	3.2	Gage
202	Cover assembly inside diameter, maximum	Level IV	3.2	Gage
203	Poor workmanship	Level IV	3.13	Visual

NOTES:

1/ See paragraph 4.5.14 for alternate method of dimensional control.

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Cover Assembly	1 OF 1	9298636
			NEXT HIGHER ASSEMBLY: 12929218

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Depth of projection diameter	Level II	3.2	Gage
Minor				
201	Diameter of small outside diameter	Level IV	3.2	Gage
202	Poor workmanship	Level IV	3.13	Visual

NOTES:

AMSMC Form 1570, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Cover and Cell Assembly	1 OF 1	12929219
			NEXT HIGHER ASSEMBLY: 12929222

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Vacuum pull test	Level II	3.2	4.5.5.2
102	Weld strength	Level II	3.2	4.4.3.3/4.5.5.1
Minor				
201	Poor workmanship	Level III	3.13	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Cell Cup	1 OF 1	12929216
			NEXT HIGHER ASSEMBLY 12929219

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Total height	Level II	3.2	Gage
102	Cell #1 Slots width	1/, Level II	3.2	Gage
103	Cell #1 Slots length	1/, Level II	3.2	Gage
104	Cell #2 Slots width	1/, Level II	3.2	Gage
105	Cell #2 Slots lenght	1/, Level II	3.1	Gage
Minor				
201	Presence of slots and holes	Level IV	3.2	Visual
202	Poor workmanship	Level IV	3.13	Visual

NOTES: 1/ See paragraph 4.5.14 for alternate method of dimensional control.

AMSMC Form 1570, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Battery, Assembly	1 OF 1	12929222
			NEXT HIGHER ASSEMBLY: 9298620

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter of case	Level II	3.2	Gage
102	Overall height	Level II	3.2	Gage
103	Case seal (pre-initiation voltage)	Level II	3.5.1	4.5.9.1
104	Activation		3.5.2	4.5.9.2/4.4.3.7.1
105	Rise time		3.5.3	4.5.9.3/4.4.3.7.1
106	Term capacity		3.5.4	4.5.9.4/4.4.3.7.2
107	Bullhead allignment, 1/	Level II	3.2	4.5.17
108	Vacuum 1/	Level II	3.2	4.5.18
Minor				
201	Poor workmanship	Level III	3.13	Visual

NOTES: 1/ 100% inspection shall be certified by process control.

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-70731A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Support, Ampule	1 OF 1	9298650
			NEXT HIGHER ASSEMBLY 12929221

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Hardness test	Level II	3.2	4.4.3.2/4.5.4
Minor				
201	Inside diameter, minimum	Level IV	3.2	Gage
202	Thickness	Level IV	3.2	Gage
203	Poor workmanship	Level IV	3.13	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Cathode. (See dwg. 9298640)

4.4.3.1.1 Moisture content. Each batch of cathode material shall be sampled for the specified moisture requirements in accordance with 4.5.2.1. If the batch passes the test requirements, it may be stored for a maximum of 90 days providing the maximum relative humidity at ambient does not exceed 5%. The batch shall be rejected if the sample is classed defective.

4.4.3.1.2 Weight. Five (5) cathodes from each station shall be weighed each production shift in accordance with 4.5.2.2. The previously made cathodes shall be rejected if one (1) or more samples are classed defective.

4.4.3.1.3 Composition. Each batch shall be tested for the specified requirement in accordance with 4.5.2.3. If the batch passes the test requirements, it may be stored for a maximum of 90 days providing the maximum relative humidity does not exceed 5%. The batch shall be rejected if the sample is classed defective.

4.4.3.2 Support ampule hardness. (See dwg. 9298650). A coupon specimen from each support ampule batch shall be made and tested in accordance with 4.5.4. The lot shall be rejected if the coupon is rejected.

4.4.3.3 Cover and cell assembly weld strength. (See dwg. 12929219). Each welder shall be monitored for control and deviation in accordance with the following procedures and as specified in 4.5.5.1.

The last six (6) samples each hour of production shall be tested for the specified weld requirement. If one (1) or more samples are classed defective, the prior hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production.

The sample size shall be reduced to three (3) samples per hour if no defects are found after 48 consecutive hours of operation. If one (1) or more samples are classed defective, the prior hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production. If two (2) consecutive samples contain one (1) or more defects, then the sampling shall revert back to the six (6) sample size category.

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The sample size may be reduced to two (2) samples every two (2) hours if there are no failures in eighty (80) consecutive hours of production. If one (1), or more samples are classed defective, the prior two (2) hours' production shall be classed defective. The welding problem shall be corrected if a defect is found during production. If two (2) consecutive samples contain one (1) or more defects, then the sampling shall revert back to the six (6) sample size category.

4.4.3.4 Cover and collar assembly weld strength. (See Dwg. 12929218). Each welder shall be monitored for control and deviation in accordance with the following procedures and as specified in 4.5.6.1.

The last six (6) samples each hour of production shall be tested for the specified weld requirement. If one (1) or more samples are classed defective, the prior hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production.

The sample size shall be reduced to three (3) samples per hour if no defects are found after 48 consecutive hours of operation. If one (1) or more samples are classed defective, the prior hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production. If two (2) consecutive samples contain one (1) or more defects, then the sampling shall revert back to the six (6) sample size category.

The sample size may be reduced to two (2) samples every two (2) hours if there are no failures in 80 consecutive hours of production. If one (1) or more samples are classed defective the prior two (2) hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production. If two (2) consecutive samples contain one (1) or more defects, then the sampling shall revert back to the six (6) sample size category.

4.4.3.5 Separator No. 1 salt content. (See Dwg. 9298331-1). Each batch of separators shall be tested for the specified requirements in accordance with 4.5.7. Three (3) samples consisting of five (5) of each disc shall be tested. The batch shall be rejected if the sample is classed defective.

4.4.3.6 Separator No. 2 salt content. (See Dwg. 9298331-2). Each batch of separators shall be tested for the specified requirements in accordance with 4.5.7. Three (3) samples consisting of five (5) of each disc shall be tested. The batch shall be rejected if the sample is classed defective.

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4.4.3.7 Battery, reserve, primary (See dwg. 12929222)

4.4.3.7.1 Activation and rise time. 125 batteries shall be randomly selected from each lot containing less than 10,000 units. The batteries shall be tested for activation in accordance with 4.5.9.2 and rise time in accordance with 4.5.9.3. The lot shall be accepted if two (2) or less are in the above sample.

Test temperatures and the number of samples per temperature group shall be in accordance with the requirements of 4.4.3.7.3

4.4.3.7.2 Capacity discharge . The batteries, having undergone successful activation and rise time shall be subjected to the specified capacity criteria. For a sample sizes of 125 units, the lot shall be rejected if there are three (3) or more defects. For sample sizes of 175 units the lot shall be rejected if there are four (4) or more defects.

Test temperature and the number of samples per temperature group shall be in accordance with the requirements of 4.4.3.7.4.

4.4.3.7.3 Test temperature and temperature groups. The samples from 4.4.3.7.1 through 4.4.3.7.3 shall be tested in accordance with Table III and Table IV.

TABLE III. Sample size for 125 batteries.

Test temperature	-25°F to ±5°F	70°F to ±5°F	125°F to ±5°F
Samples/group	42	41	42

TABLE IV. Sample size for 175 batteries.

Test temperature	-25°F to ±5°F	70°F to ±5°F	125°F to ±5°F
Samples/group	58	58	59

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4.4.3.8 Ampule weight. (See dwg. 12929220). 20 samples shall be randomly selected from each lot and tested for the specified requirements in accordance with 4.5.1.4. The lot shall be accepted if it contains no defects or rejected if it contains two (2) or more defects. In the event the lot contains one (1) defect, a second sample of 20 ampules shall be randomly selected and tested for the specified requirements in accordance with 4.5.1.4. The lot shall be rejected if the second sample contains one (1) or more defects.

4.4.3.9 Bulkhead to case force test, (See dwg. 12929220). The lot shall be tested 100% for the specified requirements in accordance with 4.5.3 when the contractor uses non-automated manufacturing techniques. Any assembly classed defective shall be removed from the lot.

The sampling shall be reduced whenever the contractor uses a fully automated manufacturing technique that does not lend itself to process interruption. For this procedure, the contractor shall select the last six (6) assemblies produced during the previous hour. The samples shall be tested for the specified requirements in accordance with 4.5.3. The quantity produced during the hour represented by the sample size shall be classed defective if one (1) or more of the samples are classed defective.

4.4.3.10 Repair. All batteries undergoing repair shall be processed in accordance with 4.5.11. Any battery classed defective shall not be returned to the lot.

4.4.3.11 Test temperature. All battery tests shall be conducted at the specified test temperature. Any battery test performed otherwise shall be classed defective.

4.4.3.12 Data requirements. All battery tests shall have the test data recorded in accordance with 4.5.12. Any material tested without the proper data acquisition shall be classed defective.

4.4.3.13 Process controls. All process controls procedures shall be processed in accordance with 4.5.13. Any process control used that is classed defective shall be cause for lot rejection.

4.4.3.14 Qualification requirements. The battery shall be inspected and tested for all the requirements specified in Table I unless specified otherwise by the technical agency listed in 6.5.

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4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

4.5 Methods of inspection.

4.5.1 Ampule

4.5.1.1 Seal. The ampule shall be isolated and completely submerged in a test solution of phenolphthalein for a minimum time of four (4) hours. The test solution shall be composed of 25 cubic centimeters (cc) of a master solution dissolved in 1000 cc of deionized water. The referenced master solution shall consist of one (1) gram of phenolphthalein dissolved in 60 cc of ethyl alcohol diluted to 100 cc with deionized water.

The ampule shall be classed defective if the test solution surrounding the ampule changes to a red or pink color. (Non-destructive test)

a. The capability of the ampule(s) to withstand exposure to the environments shall be substantiated by an evaluation of their performance or condition prior to, during, and after exposure to the environments specified in the table of inspections.

b. In addition, to the tests or examinations after the environmental exposure(s) specified in the tables of inspection, the ampules shall be visually examined to assure that they have survived the exposure(s) without physical damage that would impair their form or fit.

4.5.1.2 Temperature cycling (short term). The ampule shall be temperature cycled as specified when mounted in an isolated fixture in the temperature chamber. The ampule shall be cycled five (5) times as follows:

a. Heat soaked for a minimum of 1.5 hours at a minimum temperature of +170°F.

b. The temperature shall be gradually reduced.

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c. The ampule shall be cold soaked for a minimum of one (1) hour at a maximum temperature of -65°F.

d. This is the completion of one (1) cycle.

e. The ampule temperature shall be gradually raised and the start of the second cycle and each successive cycle shall commense. Each cycle shall be in accordance with steps "a" through "c". The five (5) cycles shall be completed within 72 hours.

Any ampule that cracks, ruptures, discolors or leaks shall be classed defective and removed from the lot. (Non-destructive test).

4.5.1.3 Ammonia composition. The ampule shall be fractured in a closed container and the vapor shall be analyzed for the specified amount of water using a gas chromatographic method or equivalent. Any ampule failing to comply with all of the specified requirements shall be classed defective. (Destructive test).

4.5.1.4 Ammonia weight. The weight shall be determined by weighing the sample ampule and then fracturing it in a closed container. The vapor shall be allowed to vent and the residue weighted. The difference in weight between the original weight and the residue shall be considered the amount of ammonia contained in the submitted sample. Any ampule failing to comply with all of the specified requirements shall be classed defective. (Destructive test).

4.5.2 Cathode

4.5.2.1 Moisture content. The sample of cathode material shall be weighted and dried at room temperature for a period not less than two (2) hours in a vacuum at a pressure of not more than 200 microns. The sample shall be reweighed and the loss shall be calculated. Any sample failing to comply with all of the specified requirements shall be classed defective. (Destructive test).

4.5.2.2 Weight. The cathode shall be weighed in an analytical balance. Any cathode failing to comply with all of the specified requirements shall be classed defective. (Non-destructive test).

4.5.2.3 Composition. A sample shall be checked by an approved quantitative analysis method for the correct percentage of materials. Any sample failing to yield the proper percentage of material shall be classed defective. (Destructive test).

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4.5.3 Bulkhead to case force test. The assembly shall be examined for the specified requirement by monitoring the specified depth at the required force. Any assembly that fails to achieve either the force or depth requirements shall be classed defective. (Non-destructive test).

4.5.4 Support ampule hardness test. The coupon shall be tested for the specified requirement using a durometer. The coupon shall be classed defective if it fails to comply with all of the specified requirements.
(Destructive test).

4.5.5 Cover and cell assembly

4.5.5.1 Weld strength. The cup and anode assembly shall be pulled apart to separation. If the parts separate without indicating a tear or a definite nugget the welded assembly shall be classed defective. (Destructive test).

4.5.5.2 Vacuum pull test. The assembly shall be secured and a vacuum probe shall be applied to the welded surfaces for a minimum period of 0.5 seconds minimum. If any plate separates during the vacuum pull, the assembly shall be classed defective.
(Non-destructive test).

4.5.6 Cover collar assembly.

4.5.6.1 Weld strength. The cover and collar assembly shall be pulled apart to separation. If the parts separate without indicating a tear or a definite nugget, the welded assembly shall be classed defective. (Destructive test).

4.5.6.2 Vacuum pull test. The assembly shall be secured and a vacuum probe shall be applied to the welded surfaces for a minimum period of 0.5 seconds minimum. If any plate separates during the vacuum pull, the assembly shall be classed defective.
(Non-destructive test).

4.5.7 Separator salt content. The separator salt content shall be checked by quantitative analysis of the disc. The analysis shall include a positive test for a presence of potassium as a major cation. The disc shall be classed defective if it fails to comply with all of the specified requirements. (Destructive test).

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4.5.8 Presence of ammonia in ampule. The ampule, prior to insertion into the case shall be visually monitored or sensed for ammonia fill. Any ampule that is cracked, low in ammonia, contains discolored ammonia, or empty shall be classed defective. (Non-destructive test). The contractor shall submit for government approval the method for testing of the minimum requirement of ammonia in the ampule.

4.5.9 Battery, primary, reserve

4.5.9.1 Case seal and pre-initiation voltage. The sealed battery shall be placed in a solvent vapor at a pressure of 100 pounds per square inch for a minimum period of 30 minutes. The vapor shall be made from the solvent of the batteries electrochemical system. Prior to five (5) hours after the battery is removed from the vapor solvent the output voltage shall be measured. Any battery indicating an output voltage in excess of the specified maximum requirement of 3.5.1 when measured with a one (1) megaohm input impedance digital voltmeter shall be classed defective and removed from the lot (Non-destructive test).

4.5.9.2 Activation. The battery shall be pre-conditioned at the applicable test temperature, placed in an approved fixed test fixture (with no spin or shaking).

Prior to activation of the battery, measure across the positive and negative leads looking for voltage. If any voltage is present the battery shall be examined for a "No Test Pre-Activation" condition.

A "No Test Pre Activation" can exist when voltage is observed across the battery leads before activation. In this case a replacement battery may be activated immediately. Test results for the replacement battery shall be used for acceptance only when analysis determines the failure is related to a cracked ampule or internal moisture causing voltage to appear before activation.

Surface "A" (See Figure 3) of the battery shall be deformed while fixed using an approved activation system.

Deformation of surface "A" shall be within twenty-five thousands (.25") inch of the X-X reference (See Figure 3). The battery surface "A" (see fig. 3) shall be deformed .25 to 0.60" inch by a spherical surface, or ball of a 7/32 +/- 1/32" inches in diameter. This test is for INFORMATION ONLY, the dent should only be measured when a battery fails to activate whether it saw the same deformation it would see in the mine. Any battery which fails to activate and does not meet the dent depth requirement will be examined for a "No Test Non-Activation".

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A "No Test Non-Activation" condition can exist when a battery fails to activate on the first try. In this case a replacement battery may be activated immediately. Test results for the replacement battery shall be used for acceptance only when an analysis determines that a "No test Non-Activation" condition exists. A "No Test Non-Activation" battery must 1) have its filled ampule intact or 2) have a nonconforming dent. In both cases all of the battery's components must be present and properly assembled. The cause of the non-activation must be attributed to the approved test equipment.

Transfer time from the preconditioning temperature chamber to the test fixture shall be such that the test sample temperature shall not vary more than 10°F. Any battery failing to comply with all of the specified requirements shall be classed defective. (Destructive test)

4.5.9.3 Rise time. The battery output voltage shall be continuously monitored as follows:

a. Continuously for the total test time at Hot and Ambient.

b. At a rate of at least once every ten (10) seconds at Cold temperture.

c. Across a 3300 ± 5 percent OHM load resistor.

The battery shall be classed defective if it fails to comply with all of the specified requirements. The sample shall be activated in accordance with the method specified in 4.5.10.2. (Destructive test).

4.5.9.4 Time capacity. The battery shall be activated in accordance with 4.5.9.2, and measured for its rise time in accordance with 4.5.9.3. The battery output voltage under load shall be continuously monitored on an approved date acquisition system for a minimum of four (4) hours. The battery shall be classed defective if it fails to comply with all of the specified requirements. (Destructive test).

4.5.10 Environmental and mechanical tests

4.5.10.1 Thermal shock. The battery, in an approved test vehicle, shall be tested in accordance with Test Method 113 of MIL-STD-331. The battery shall not activate during test but be capable of normal operation in accordance with 4.5.9.4. The battery shall be classed defective if it fails to comply with all of the specified requirements. (Non-destructive test).

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4.5.10.2 Temperature cycling. The battery, in an approved test vehicle, shall be tested in accordance with Test Method 105 of MIL-STD-331. The battery shall not activate during test but be capable of normal operation in accordance with 4.5.9.4. The battery shall be classed defective if it fails to comply with all of the specified requirements. Note the humidity requirements are not required for testing.

4.5.10.3 Transportation vibration. The battery shall be tested in accordance with Procedure I, Test Method 119 of MIL-STD-331 shall be used except test vehicle shall be randomly located in the standard shipping package. The battery shall not activate during test but be capable of normal operation in accordance with 4.5.9.4. The battery shall be classed defective if it fails to comply with all of the specified requirements.

(Non-destructive test). Test temperature groups shall be -52°C ± 2°C 30°C + 10°C and +60°C ± 2°C.

4.5.10.4 Drop test. The battery, in an approved test vehicle, shall be tested in accordance with Test Method 111 of MIL-STD-331 except the height of drop shall be seven (7) feet. The battery shall not activate during test but be capable of normal operation in accordance with 4.5.9.4. The battery shall be classed defective if it fails to comply with all of the specified requirements. (Non-destructive test).

4.5.11 Repair. Any battery undergoing repair shall have the procedures submitted to the technical agency listed in 6.5 prior to performing the necessary work. Any battery repaired without prior Government approval shall be considered as non-conforming material and classed defective (Non-destructive procedure).

4.5.12 Data requirements. The contractor shall submit a quality inspection test, demonstration and evaluation report detail in accordance with Data Item Description DI-QCIC-81187 (tailored) giving all detailed inspection and test results for test performed as required by 4.4.3.7. Test report format shall be submitted to the technical agency listed in 6.5.

4.5.13 Process controls. All essential process control procedures shall be documented and submitted to the technical agency listed in 6.5. Any battery component, sub-assembly or assembly undergoing a process control deemed necessary to assure a reliable end item by the Government that has not been previously submitted to the Government shall be classed as non-conforming material and classed defective. All changes to these process controls must be submitted to the Government prior to the incorporation of the process control.

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4.5.14 Dimensional control of molded or plastic parts. In place of the normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (See Note 1) a sample of three (3) parts (as molded) from each cavity shall be fully inspected dimensionally to quality a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in 4.4.2.1 through 4.4.2.13 after continuous production of each 5000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect.

All cavities removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request. Contractor designs of gages and test equipment required to perform the inspections listed herein shall be forwarded to the governments technical agency for approval prior to manufacture of equipment. The noted sub-paragraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60 minutes etc.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to the governments technical agency. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

4.5.15 Dimensional control of stamped parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

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If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

4.5.16 Ammonia weight screen. Each PDM ampule shall be screened utilizing processes and procedures approved by the Government. Any ampule failing to comply with the requirements of 3.3.4 shall be classed defective and removed from the lot. (Non-destructive test)

4.5.17 Bulkhead alignment. The contractor shall use a Government approved process for positioning the bulkhead into alignment with the cell cup as shown in dwg. 12929222. Any battery or production group of batteries not complying with this criteria shall be classed defective and removed from the lot. (Non-destructive process)

4.5.18 Vacuum. The battery prior to welding shall comply with the vacuum requirements per dwg. 12929222, using a Government approved process. Any battery failing to comply with the specified criteria shall be classed defective and removed from the lot. (Non-destructive test)

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

5. PACKAGING

5.1 Preservation and packaging

5.1.1 Level C. The batteries shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing

5.2.1 Level C. The batteries preserved and packaged as specified in 5.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking. Marking shall comply with MIL-STD-1169.

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5.4 Shipping. When shipments from more than one lot are shipped, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The components and assembly covered by this specification are intended for use as the power supply for molded assembly for the Mine, Anti-Personnel, M86.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of indivisual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract, per the Contract Data Requirements List, DD Form 1423.

6.4 Submission of inspection equipment designs for approval. (See MIL-A-48078) Submit equipment designs as required to: Commander, ARDEC, ATTN: SMCAR-QAT-I, Picatinny Arsenal, New Jersey 07806-5000. This address will be specified in the Contract Data Requirements list, DD Form 1423 in the contract.

6.5 Submission of alternative quality conformance provisions. All contractor, proposed alterntive quality conformance provisions will be submitted to the Government for evaluation and approval as directed by the contracting activity.

6.6 Distribution of ammunition data cards, inspection and test results, process controls. Copies of ammunition data cards, inspection and test results, process controls shall be sent to: Commander, ARDEC, ATTN: SMCAR-QAT-M, Picatinny Arsenal, New Jersey 07806-5000

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development Engineering Center, (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.8 Subject term (key word) listing.
Electrolyte
Voltage Rise Times

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project: 1345-A621)

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